

Abu Bakar

Software Developer

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Computer Science undergraduate focused on AI-assisted software systems, machine learning, and full-stack development. Skilled in Python, Flask, React, Next.js, and the MERN stack, with knowledge of software design, UML diagrams, and SRS documentation. Experienced in integrating ML and LLM-based features into real-world applications, with exposure to NLP, data analysis, and cloud deployment.

EDUCATION

BS in Computer Science 10/2022 – 07/2026
Namal University Mianwali

FYP: SudhaarAI – Generative Code Assistant and Context-Based Code Optimizer

Relevant Courses: Artificial Intelligence, Machine Learning, Natural Language Processing, Cloud Computing, Web Application Development

PROJECTS

SudhaarAI (FYP)

Built a full-stack AI code optimization platform that analyzes Python projects, retrieves project-specific context through a RAG pipeline, and generates code improvement suggestions using multiple LLM providers.

Implemented multi-turn chat, Monaco Editor code diffing, vector-based project knowledge ingestion, Clerk JWT authentication, MongoDB session storage, and automated LLM fallback handling.

Cotton Seed Classifier

Developed a web and Flutter-based cotton seed classification system using a ResNet50 CNN model to classify 7 cotton seed varieties.

Integrated batch prediction visualization with Chart.js to make model outputs easier to review for agricultural and academic evaluation.

BarberAI

Built and deployed an AI-powered barber booking platform on Google Cloud Run using Gemini 2.5 Flash Vision.

Implemented face-image analysis and personalized hairstyle recommendation flow to support user-specific appointment decisions.

Ustad

Built an AI automation tool that converts Google Cloud Skills Boost lab instructions into executable Bash scripts using Gemini.

Developed progress tracking, Markdown export, Clerk authentication, and PostgreSQL/Prisma-backed data persistence in a Next.js application.

MyNamalGPA

Developed a GPA/CGPA calculator for Namal University students using React, TypeScript, and shadcn/ui. Added HEC grading logic, course repeat handling, PDF export, and basic client-side security controls.

SKILLS

Python JavaScript TypeScript SQL C++

Machine Learning Deep Learning NLP RAG

LLM APIs Scikit-learn PyTorch TensorFlow

React.js Next.js Flask FastAPI REST APIs

MongoDB ChromaDB MySQL Neo4j

Git GitHub Microsoft Azure Google Cloud Run

PUBLICATIONS

Data-Driven Analysis and Predictive Modeling of Urban Air Quality for Environmental Management

Authored by: M. Raqib Hayat, Abu Bakar, M. Bilal, and M. Ramzan Shahid Khan.

Published in *International Journal of Innovations in Science & Technology*, Special Issue, pp. 508–527, May 2026. 

CERTIFICATIONS

Machine Learning Fellowship — ByteWise Limited

AI for Everyone — DeepLearning.AI

AI Fundamentals — DataCamp

Programming for Everybody — University of Michigan

ACHIEVEMENTS

2nd Runner-Up, Agri Fusion 2025: Recognized among 35 FYP projects for the Cotton Seed Classifier.

Beta Microsoft Learn Student Ambassador: Representing Microsoft Learn and supporting student tech communities.

Google Cloud AI Study Jam: Completed 31 Google Cloud skill badges in AI and cloud tracks.

Google Cloud AI Seekho 2026: Completed AI tracks and built BarberAI using Google AI Studio.

Hacktoberfest 2024: Completed open-source contribution challenge with accepted pull requests.

Head of Skill Development Wing, NSSI Society: Led university students in teaching technical skills to local namal students.